Bike Sharing Analysis Dashboard

1. Project Overview

The Bike Sharing Analysis project focuses on analyzing station-level data from a public bike-sharing system operating in multiple cities, mainly across Europe. The project studies station availability, bike and stand capacity, station status, banking (payment terminal) availability, and city-wise distribution. Excel is used for data cleaning and preliminary analysis, while Power BI is used to build an interactive dashboard. The project transforms raw station data into insights that support better planning of bike-sharing operations.

2. Problem Statement

A bike-sharing operator manages thousands of stations spread across many cities. Each station reports how many bikes and free stands it has, whether it is open or closed, and whether it offers banking (card payment). Reviewing this data manually makes it hard to see which cities have the most capacity, where bikes are running short, how many stations are out of service, and how stations are spread geographically. The lack of an integrated view limits the ability to rebalance bikes, plan maintenance and decide where to expand. Therefore, this project aims to clean, analyze and visualize the bike sharing dataset using Microsoft Excel and Power BI to identify meaningful insights and support data-driven operational decisions.

3. Attribute (Column / Features) Details

Attribute Name	Data Type	Description
Station ID	Integer	Unique identifier of each bike station record.
Station Name	Text	Name of the station.
Address	Text	Street address / location description of the station.
City	Text	City (contract area) in which the station is located.
Latitude	Decimal	Latitude coordinate of the station.
Longitude	Decimal	Longitude coordinate of the station.

Station Status	Text	Whether the station is OPEN or CLOSED.
Banking Available	Boolean	Whether the station has a payment terminal (True / False).
bonus	Boolean	Indicates whether the station is a bonus station.
Total Bike Stands	Integer	Total number of bike stands (docks) at the station.
Available Bike Stands	Integer	Number of empty stands currently available for returning bikes.
Available Bikes	Integer	Number of bikes currently available for rent.
Last Update Date	Date	Date on which the station data was last updated.
Last Update Time	Time	Time at which the station data was last updated.

4. Tools & Technologies

- **Excel:** Data cleaning, transformation and Pivot Tables.
- **Power BI:** Data modelling, DAX calculations, visualization and interactive dashboard creation.

5. Data Modelling and DAX (Power BI)

- **Data Model:** The dataset is loaded as a single flat table named **bike**. Since all station attributes are stored in one table, no additional relationships were required.
- **Calculated Columns & DAX Measures:**

Total Stations (Count of Station ID)

```
Total Stations = COUNT(bike[Station ID])
```

Total Available Bikes

```
Total Available Bikes = SUM(bike[Available Bikes])
```

Total Bike Stands

```
Total Bike Stands = SUM(bike[Total Bike Stands])
```

Available Bike Stands

```
Available Bike Stands = SUM(bike[Available Bike Stands])
```

Active Stations

Active Stations = CALCULATE(DISTINCTCOUNT(bike[Station ID]), bike[Station Status] = "OPEN")

